

ChE 363- Separation Processes and Mass Transfer Spr 25 Unique: 16130

Recitation: Mandatory. Aspen software sessions will be in recitation only. Fri 1-2 UTC 4.104

Required Text: McCabe, Smith and Harriott "Unit Operations of Chemical Engineering" Sixth or Seventh edition (These editions are very similar). Electronic rental of the book is less expensive and just fine. Pirated pdfs not allowed as it is stealing of intellectual property.

Related references in chemistry library:

Seader, J.D., Henley, "Separation Process Principles" (very extensive)

Cussler "Diffusion: Mass Transfer in Fluid Systems " (non-required suppl. text – key info in sup. Notes in Canvas).

Prof: Keith Johnston, Office Hours: CPE 5.414 hybrid incl. zoom M 1 and W 4:30

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TA: Tsung Lun Lee Office Hours: M and W: noon to 1 CPE 2.222

Prerequisites: ChE 322 Thermo, ChE 319 Transport

Chapters (McCabe et al. (6th or 7th edition required- very similar editions) (see calendar in files)

1. 17 Diffusion and Mass Transfer between Phases (steady and unsteady state) (also Cussler) ~3 ½ weeks
2. 20 Equilibrium stage operations 1/2
23 Extraction including Bioseparations 1/2 (we will limit this to dilute systems)
3. 18 Gas Absorption (excluding rich gases and chemical reaction) 1.5
4. 21 Distillation 2
22 Multicomponent distillation ½ (not on exams, also covered in recitation with Aspen software)
5. 25 Fixed-bed separations: Adsorption (primarily), chromatography, ion exchange, materials chemistry 2
6. 26 Membrane separations; gases, liquids- osmosis and reverse osmosis 2
7. 27 Crystallization/Nucleation/Nanotechnology 1.5

Grading method: 3 Hourly exams (60%), Final exam (30%), HW and quizzes (10%).

Public health: Getting vaccinated for covid/flu reduces the chance of infection of the individual and everyone they come in contact with. Getting tested and staying home if you have symptoms or test positive is recommended. For more info: protect.utexas.edu.

Exams: In class. **Th 2/8 , T 3/19, Th 4/11 in class.** Final exam: **F. May 3 1-3.** Closed book.

One sheet of notes front side only. Mass transfer summary sheet provided. Pls give me advance notice if you are not able to attend an exam.

Homework and recitation quizzes: Homework will be assigned officially under assignments in Canvas for most classes to encourage continuous learning. Upload into assignments on Canvas. The preliminary unofficial HW listing is in the file: 363 spring 24 calendar.

Lecture start time: Pls arrive a few minutes early so you are ready to go. There is no need to look at the asynchronous lectures. I only included them since a minority of students like to have them.

Coordination of primary power point notes, supplemental notes, learnchemE screencasts and text: Pls see documents on Canvas in Files under various chapters..

Homework and examples: Canvas has a table that explains homework by type and how long it takes. The goal is to help you to classify types of problems, as this will help you prepare for tests.

Canvas: "Files" includes general info as well as material specific to each chapter as filed under the folder Chapters.

LearnChemE screencasts: <http://www.learncheme.com>. See separations/mass transfer screencasts All of you learn differently. We wish to reach everyone. These highly acclaimed optional screencasts for PCs to tablets to phones may be very helpful to many of you.

Abilities before entering

Understanding of vapor-liquid equilibrium and liquid-liquid eq. Mass and energy balances, Fick's laws and mass transfer section of ChE 319.

Use of spread sheets which will be used to understand solutions of stage wise problems.

Abilities students should gain

Outcome: Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

Engineering design for open ended problems: How to synthesize material balances, thermodynamics and mass transfer to do stagewise calculations- often requires engineering assumptions

Solution of stage wise mass transfer and equilibrium problems
spreadsheets- to learn concepts of McCabe-Thiele graphical solutions
process simulator (e.g. Aspen)

Thermodynamic calculations in distillation and stage efficiencies

Understand key components and concentration profiles in multicomponent distillation

Mass transfer coefficients, correlations and film theory (absorption, adsorption etc.)

Partial differential equations: mass transfer and adsorption chapters

Molecular concepts and materials chemistry for achieving selectivity in separations

Nucleation, growth and crystallization (examples given for nanotechnology)

Consideration of safety, energy utilization, sustainable technologies, pollution, emerging technologies and the role of government policy and society

Application to broad range of industries: energy, petrochemicals, environmental, nanotechnology, biotechnology, pharmaceuticals-drug delivery, transportation (fuel cells)

Tips to maximize success:

1. It is an honor to have you in our department and in my class as you have already excelled. Continue to develop a passion for knowledge and life-long learning!
2. Class etiquette and professionalism- maximize learning by giving the course your full attention during class..
3. Review info where you first learned it. Reread key sections in calculus book, transport book, thermo book etc. and your old notes. The more you review these topics the better you will retain it.
4. The textbook is a key element for this class. As you do lifelong learning, reading books and articles play an important role. **The notes are not meant to be self-sufficient, but to complement the text. Read them together!!**
5. See us in office hours. Participate in lecture and recitation. Students often chat with me right after class as well.
6. Find other students that study together online.
7. **It usually takes ~ 8 hours/week outside of class to read the book including examples in chapters (in conjunction with notes), learn the theory and concepts, and to do the homework problems.**
8. Strive to understand the theory and concepts in addition to doing problems.

9. Learn to identify classes of problems based on theoretical concepts. The summary sheets called exam study guide in the chapters will help here.
10. Maximize what you get out of the lectures by keeping up and understanding all concepts after each lecture. If you get behind, try to immediately catch up to max. efficiency for learning all classes each week.
11. Look ahead in the book and notes for about 5-10 min before each lecture. The brain will then be activated and eager to learn very efficiently and deeply.
12. Avoid the short cut to use old homework solutions as a crutch. It is far easier to follow somebody else's solution than to construct your own.
13. Students should do all homework problems and study solutions after turning in homework as needed on Canvas. Save hws when you get them back so you can study your approach vs. posted solutions.
14. Always check units in eqns. They are important. Check with factor label. This approach helps with exam grades.